



Fajar Aziz Laksono

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Summary

Research-oriented Software Engineer and Business Analytics Master's candidate with 7+ years of experience in system design, data analysis, machine learning implementation, and large-scale backend systems. Strong interest in Artificial Intelligence, applied machine learning, data-driven optimization, and technology impact research. Experienced in experimental design, performance benchmarking, automation research, and cost optimization modeling.

Research Interest

Software Engineering, • Data Engineering & Scalable Systems, • Applied Machine Learning & Predictive Modeling, • Optimization & Performance Benchmarking, • AI in Industry & Economic Impact of AI, • Automation Systems & DevOps Research, • Business Analytics & Data-Driven Strategy.

Education

Master of Business Analytics, Universiti Sains Malaysia, Malaysia. 2024-2026

- GPA: 3.58/4.00.
- Relevant Coursework: Data Programming, Predictive Analytics for Business, Machine Learning, and Economic and Management Insight.
- Organization Involvement: AI Trainer in Indonesian Students Association
- Publications: Predictive Analysis of Cloud Resource Utilization and Cost Optimization Using Machine Learning on Azure VM Traces (2026)

Bachelor of Information System Management, Universitas Bina Sarana Informatika, Indonesia. 2015-2019

- GPA: 3.79/4.00.
- Relevant Coursework: Databases and Data Structures, Software Engineering, and Project Management.
- Organization Involvement: Leader and Teacher in Programming and Animation Forum
- Publications: Analysis of YouTube User Satisfaction as an Educational Medium Using the DeLone and McLean Model (2019), • Learning Management System Web-Based Application for Computer Science (2018).

Professional Experience

Shieldbase AI - Malaysia, Singapore-based

Technical Product Lead, March 2025 - April 2026

- Led cross-functional teams across Product, Engineering, and QA, defining roadmap priorities and execution plans while maintaining delivery predictability and quality standards.
- Translated business and customer requirements into clear technical specifications, backlog items, and release plans, enabling efficient and consistent product delivery. System design and technical decision-making evaluation for AI-based solutions, balancing scalability, performance, and long-term maintainability.

AccelByte Inc - Indonesia, USA-based

Analytics Engineer, March 2021 - December 2024

- Led a research initiative to identify and implement strategies for reducing CDN and Storage costs, leveraging data analysis, benchmarking, and evaluating alternative providers. Ultimately, I successfully introduced an optimized solution, onboarded the new vendors, realized around 40% reduction in CDN and Storage costs, and the research was adapted to all company products. Significantly improving operational efficiency and cost savings.

- Led a research initiative to automate the release and Deployment of the services by implementing an Automated Release and Deployment design. As a result, it has successfully improved team productivity speed during the release train by 25%-40% and reduced manual intervention.
- Led an initiative to automate data warehouse migration script by implementing Flyway and made a sandbox environment, successfully improving Database Engineers' productivity by 25%-40% and reducing manual migration errors.
- Research, designed and developed scalable and stable backend solutions by collaborating with cross-functional teams and implementing system designs, ensuring on-time delivery of features and enhancing user satisfaction

Agate International - Indonesia

Full-stack Programmer, September 2020 - March 2021

- Designed and developed the backend system and its admin portal for a game, ensuring a robust and scalable infrastructure that supported seamless gameplay and management features. Ensured timely delivery of the project by adhering to deadlines, coordinating with cross-functional teams, and mitigating obstacles to maintain the project schedule.

PT. Gria Inovasi Teknologi - Indonesia

Software Engineer, September 2019 - March 2020

- Designed and Developed backend and frontend for a national management system for the Attorney General of Indonesia using PHP Laravel and Bootstrap. Coordinated daily with stakeholders to refine requirements and delivered the system on schedule with a user-friendly interface and efficient server-side logic.

Universitas Bina Sarana Informatika - Indonesia

Lab Assistant, June 2016 - September 2016

- Assisted in laboratory instruction for programming courses (PHP, C/C++, Visual Basic). Guided students in debugging, system logic design, and structured problem-solving. Supported academic instruction and practical experimentation sessions.

PT. Mitra Mandiri Informatika - Indonesia

Junior Programmer, April 2016 - September 2016

- Contributed to building decision support system (DSS) applications. Gained hands-on experience in programming and application architecture. Employed VB.net + Microsoft Access and PHP + MySQL technologies.

Portfolio

Azure Cloud Predictive Analysis • AI-Neuroscience: EEG-Based Parkinson's Disease Classification • Heart Disease Analysis • Dirty IMDb Top 1000 Movies Data Cleaning With LLM Gemini • Eye Health Analysis and Machine Learning Models Selection • and **See more portfolios:** <http://fajarlaksono.github.io/research>

Technology Experience

- **AI & Agentic Systems:** CrewAI, LangChain, LiteLLM, mem0ai, ollama, OpenCode, OpenClaude, Hugging Face Transformers, Sentence Transformers, MiniLM.
- **Data Architecture & ETL:** Databricks, Snowflake, DuckDB, Kafka Connect, Apache Airflow, Milvus, Vector PostgreSQL (pgvector), MySQL, MongoDB, Elasticsearch, Redis
- **Software Engineering:** Go, Python, C/C++, C#, Spring Boot, FastAPI, Next.js, Laravel.
- **Data Science & Machine Learning:** PyTorch, TensorFlow, Scikit-learn, Pandas, NumPy, PyArrow, Joblib.
- **Cloud, DevOps & IaC:** AWS, GCP, Azure, Docker, Kubernetes, Terraform, GitHub/GitLab CI/CD, Makefile.
- **Data Visualization & XAI:** Power BI, Looker, Grafana, Jupyter Notebook, Seaborn, Matplotlib, SHAP.
- **Product & Design Collaboration:** Figma, Jira, Linear, Confluence, Miro.